Dan Hlevca

914-501-7617 | dan.hlevca@gmail.com | danhlv.github.io | Playa del Rey, CA 90293 |
U.S. Citizen — Eligible to obtain and maintain a Security Clearance

Summary

Multi-physics engineer with 15+ years of experience spanning aerodynamics, thermal-fluid systems, propulsion, and automotive applications. Expert in reduced-order modeling (POD, SPOD, DMD), physics-informed ML, surrogate modeling, and AI-driven design-space exploration. Proven track record leading cross-functional technical teams and translating advanced AI/ML methods into engineering environments. Combines deep physics and simulation expertise with rapidly evolving generative AI and computational methods to redefine optimization, accelerate scientific discovery, and enable next-generation engineering design.

Ph.D. Fluid Mechanics - Passive & Active Flow Control Methods | **M.S.** Data Science (ML/AI)

Technical Skills

Multi-Physics Simulation: CFD (OpenFOAM, ANSYS Fluent), thermal analysis, heat transfer, fluid-structure interaction.

AI/ML for Engineering: Surrogate modeling, physics-informed neural networks, reduced-order models (POD, SPOD, DMD), neural operators, generative design, inverse design

Engineering AI Platforms: Physics AI workflows, simulation-in-the-loop ML, automated design optimization, parametric geometry generation, AI-driven engineering automation

Programming & Automation: Python (NumPy, SciPy, PyTorch, TensorFlow), MATLAB, Siemens NXOpen API, ANSYS scripting, pipeline automation, batch processing

Data Engineering: data pipelines, ETL workflows, large-scale data processing, graph databases

AI-Native Workflow: Daily use of agentic coding tools (Codex, ChatGPT) for simulation automation, data analysis, and engineering workflow optimization

Experience

Senior Mechanical Engineer, Systems Thermal & Fluid Dynamics June 2026 – Present
CesiumAstro - Communication Systems for Space, Air and Ground *El Segundo, CA*

- Created a 10-phase Python + Siemens NXOpen pipeline converting controlled thrust-chamber profiles into a reusable regenerative-cooling CAD model with 136 LOX channels and 242-to-484 kerosene bifurcation branching networks
- Generated production-ready model with 6,260 curves and 741 rib surfaces; passed 24,654 automated geometry/contract checks in 32.4 seconds
- Automated Siemens NX construction + full verification: 100% pass rates (5/5 immediate, 12/12 envelope, 12/12 layered deliverable checks)
- Built independent additive manufacturing reviewer with strict process validation, deterministic decision gates, NX-based overhang measurements, evidence hashing, and automated remediation reports
- Diagnosed and fixed spline geometry oscillation caused by clustered design stations, preserving original data and documenting cleanup process
- Optimized propulsion components for 3D printing: replaced rounded LOX outlets with 45-degree triangular features, added paired kerosene U-turns, modified common manifold for flow continuity + manufacturing compatibility
- Built and maintained 83-test automated verification suite covering configuration, profile handling, topology, CAD contracts, production generation, printability, decision tracking, and reporting
- Developed an API-driven engineering platform that centralizes rocket engine components, schematics, functional capabilities, and simulation tools, enabling unified access and interaction across engineering workflows
- Integrated geometry, operating parameters, and simulation outputs into a unified workflow enabling rapid evaluation and iterative propulsion-system design

- Developed a production-grade Python simulator for a dual-fluid regenerative-cooling system, spatially resolving coupled energy and momentum balances across changing LOX passage topology and multi-pass RP-1 routing to predict pressure drop, coolant temperature/velocity, heat absorption, phase/Mach behavior, and wall-temperature profiles
- Implemented real-fluid LOX thermofluid modeling using CoolProp and relaxed fixed-point iteration at each axial segment, resolving enthalpy-dependent temperature, density, velocity, friction, acceleration, and local-loss effects to a 0.5 Pa momentum-residual tolerance, with automated warning and hard-failure safeguards
- Developed coupled coolant–wall thermal modeling incorporating Reynolds/Prandtl-dependent correlations, Gnielinski heat transfer, fin-efficiency-corrected wetted area, through-wall conduction, and axial wall conduction to predict coolant- and hot-gas-side wall-temperature distributions
- Built a reproducible Monte Carlo uncertainty-quantification and global sensitivity-analysis framework, executing the production solver across 512 seeded samples and 10 uncertain operating/model inputs using truncated-normal and triangular distributions to quantify uncertainty propagation, establish P05/P50/P95 performance envelopes, and identify the dominant parameters governing thermal and hydraulic performance

Lead Mechanical & Simulation Engineer June 2024 – January 2025

MATTUR - Powering a Better Future *Scottsdale, AZ*

- Built parametric CAD models and automated geometry workflows for iterative design exploration of shrouded wind turbine assemblies
- Performed simulation-in-the-loop design optimization: parametric geometry sweeps evaluated through CFD, with results feeding back into design modifications
- Applied linear and nonlinear optimization methods to geometry design, trading shape parameters against performance and packaging constraints
- Developed Python-based automation scripts for batch geometry processing, simulation execution, and post-processing of design variants
- Created simulation-ready meshed models using ANSYS Geometry and ANSYS Mesh for parametric design evaluation
- Validated simulation results against experimental data, calibrating models for reliable design-space exploration
- Generated technical documentation including parametric component specifications, design trade studies, and optimization reports

Senior Fluid Simulation Engineer January 2023 – May 2024

SMOORE International *Scottsdale, AZ*

- Built automated simulation pipelines using ANSYS Fluent and OpenFOAM for batch evaluation of parametric design variants
- Developed meshing workflows for complex 3D assemblies, optimizing mesh quality for simulation accuracy and throughput
- Applied reduced-order modeling techniques for rapid design-space exploration across geometric parameter sweeps
- Created Python-based automation scripts for geometry manipulation, simulation execution, and post-processing of results
- Designed experimental validation setups to calibrate simulation models for reliable parametric design evaluation
- Generated simulation-ready datasets from parametric CAD models, supporting data-driven design optimization
- Produced technical documentation for automated workflows, simulation methodologies, and design exploration results

Research Engineer - Numerical Aerodynamics and Fluid Mechanics Nov. 2016 – July 2022

DynFluid Laboratory - Arts et Metiers ParisTech *Paris, France*

- Developed automated simulation pipelines: meshing strategies, solver configuration, and post-processing workflows for parametric design evaluation of aerospace and automotive geometries
- Created numerical methods and mesh generation workflows for complex 3D geometries, optimizing mesh quality for simulation accuracy
- Built reduced-order models (POD, SPOD, DMD) for rapid prediction across geometric parameter sweeps, enabling high-throughput design exploration
- Applied simulation-in-the-loop design methodology: parametric geometry variants evaluated through CFD with results informing design modifications
- Produced peer-reviewed technical documentation, simulation methodologies, and design exploration reports for multi-year programs

- Collaborated with distributed international research teams, aligning on simulation standards and validation workflows
- Published peer-reviewed journal articles in AIAA, Experiments in Fluids, and European Journal of Mechanics/B Fluids
- Optimized sensor placement and measurement strategies for experimental data acquisition and simulation validation

Research Engineer - Experimental Aerodynamics and Fluid Mechanics March 2014 – Oct 2016

Institut AeroTechnique *Saint Cyr l'Ecole, France*

- Planned and executed experimental test campaigns: test matrix design, instrumentation strategy, sensor selection (thermocouples, pressure transducers, flow meters), and data acquisition
- Performed thermal-fluid testing using PIV, Kulite sensors, Pitot tubes, Kiel probes, and aerodynamic balance systems
- Coordinated implementation, qualification, and validation of measurement systems and instrumentation for experimental campaigns
- Correlated CFD simulations with experimental data, calibrating computational models against measured system performance
- Managed experimental data processing, statistical analysis, and data reduction into system performance models
- Identified and supported component-level and sub-system testing required for risk mitigation
- Developed test analysis reports and recommended design improvements based on experimental findings

Contractual Professor September 2014 – Oct. 2016

EPF Engineering School *Sceaux, France*

- Designed and delivered graduate-level curriculum in Numerical Methods in Fluid Mechanics for Aeronautics & Space master students, distilling complex CFD concepts for interdisciplinary audiences
- Created 2D and 3D geometry and mesh generation workflows using ANSYS, producing educational materials and technical documentation for student reference
- Developed problem-solving sessions, term projects, and assessments requiring students to articulate technical concepts through written reports and presentations
- Postprocessed simulation results with Python and TecPlot, creating data visualization materials for classroom instruction and knowledge transfer

Assistant Professor September 2009 – March 2014

Technical University of Engineering *Bucharest, Romania*

- Conducted professional development seminars for engineers on wind engineering and hydraulic machinery, distilling complex aerodynamic concepts for practicing professionals
- Designed and led experimental research programs, creating technical documentation and standard operating procedures for wind tunnel facilities
- Instructed senior-level fluid mechanics courses, developing assessments and providing detailed written feedback on student technical reports
- Lectured on laboratory techniques and theoretical framework, translating complex fluid mechanics principles into actionable knowledge for students
- Integrated wind tunnel instrumentation with data acquisition software, optimizing measurement workflows for experimental reliability

Education

M.S. in Data Science – ML/AI Focus Tempe, Arizona

Arizona State University

Coursework: *Machine Learning, Artificial Intelligence, Data-Driven Modeling, Computational Methods for Engineering Systems, Applied Statistics*

Focus: *Application of ML/AI techniques to mechanical and aerospace systems, including fluid dynamics, control systems, and simulation-based optimization.*

Graduated: *May 2026*

Ph.D. Aerodynamics & Flow Control Bucharest, Romania

Technical University of Engineering | Joint French/Romanian European Research Grant *Sep. 2009 – Aug. 2012*

Thesis: Numerical and experimental research on flow control in wind tunnels

B.S./M.S. in Mechanical and Aerospace Engineering Sep. 2004 – Aug. 2009

Aerodynamics, Fluid Mechanics, Thermal Engineering

AI/ML Engineering Projects

AeroML: Generative AI for Airfoil Design | *generative modeling, inverse design, aerodynamic optimization*

- Developed generative AI pipeline for airfoil geometry optimization using autoencoders and inverse neural networks
- Trained autoencoder on NACA airfoil geometries to capture shape distribution in latent space
- Built inverse neural network mapping aerodynamic performance parameters to latent geometries, enabling physics-informed generative design
- Implemented end-to-end pipeline: performance specifications → inverse NN → latent geometry → decoder → optimized airfoil shape

AI-Based Wind Turbine Pitch Control | *neural networks, real-time control, renewable energy*

- Developed neural network-based pitch control system for wind turbine blades adapting to incoming wind conditions
- Implemented real-time optimization maintaining optimal rotational speed across varying wind profiles
- Applied physics-informed ML approach combining aerodynamic models with data-driven control strategies

AI Engine Control Unit | *embedded AI, real-time systems, combustion optimization*

- Developed AI algorithm governing generator engine functioning using neural network-based control
- Created real-time inference system for combustion optimization and performance monitoring

Neo4j + Kafka Data Pipeline | *data engineering, graph analysis, scalable systems*

- Built data ingestion pipeline with Kafka Connect and Neo4j for large-scale graph analysis
- Applied to NYC taxi data analysis demonstrating scalable data engineering workflows

RAG System for Engineering Documents | *retrieval-augmented generation, NLP, knowledge systems*

- Built RAG system using SentenceTransformers for embeddings, FAISS for retrieval, and GPT-4 for generation
- Implemented document question-answering for engineering knowledge extraction

Wind Tunnel Roughness Surrogate Model | *PCA-GPR, inverse design, Gradio deployment*

- Built PCA-GPR surrogate models predicting aerodynamic profiles (velocity + turbulence intensity) from 14 roughness element configurations in wind tunnel
- Implemented inverse design solver finding roughness configurations producing desired velocity profiles using L-BFGS-B optimization
- Deployed interactive Gradio web application for real-time inverse design with alpha shape parameter control
- Achieved $R^2 \approx 0.8+$ for forward prediction across 170 wind tunnel configurations

Solar Panel Shedding Frequency Calculator | *vortex shedding, resonance analysis, Streamlit dashboard*

- Built standalone desktop application analyzing vortex shedding frequencies and resonance risks in solar panel installations
- Implemented f/U ratio calculation with interactive tilt angle analysis and resonance detection with configurable tolerance bands
- Deployed Streamlit dashboard with Plotly visualizations, CSV export, and real-time parameter exploration
- Delivered production-ready executable for engineering team with automated browser launch and local server

Languages

English: full professional proficiency
French: full professional proficiency
Spanish: limited professional proficiency
Romanian: native language

Publications & Peer Review

1. Modal analysis of actively controlled flow past a backward facing ramp - (2020) AIAA - https://arc.aiaa.org/doi/10.2514/6.2020-0100

2. Atmospheric Boundary Layer Modeling in a Short Wind Tunnel - Europeean Journal of Mechanics/B fluids (2020) - https://doi.org/10.1016/j.euromechflu.2019.10.003

3. Experimental study of the active control applied to the flow past a backward facing ramp - Experiments in Fluids (2018) - https://doi.org/10.1007/s00348-018-2497-0

4. Active Control applied to the flow past a backward facing ramp: S-PIV measurements and POD analysis - (2015) 3AF France - https://www.3af-aerodynamics.com

5. Atmospheric Boundary Layer Modeling in a Short Wind Tunnel - (2014) 3AF France - https://www.3af-aerodynamics.com

6. Numerical predictions of the flow around a profiled casing equipped with passive flow control devices - (2013) Journal of Wind Engineering and Industrial Aerodynamics - http://dx.doi.org/10.1016/j.jweia.2012.12.006

7. Penstock failure detection system at the "VÂLSAN" hydro power plant - (2012) - 26th IAHR Symposium on Hydraulic Machinery and Systems, Tsinghua University, Beijing, China

8. Device for passive flow control around vertical axis marine turbine - (2012) - 26th IAHR Symposium on Hydraulic Machinery and Systems, Tsinghua University, Beijing, China

Granted patent

1. Annular housing optimized for horizontal axis wind turbines - https://EuropeanPatentOffice.org